Caylie McMahon, Sam Gold, Oscar de Castro, Lakhvir Sohal

Prof. Fredrickson

Exploring the Coefficient of Variation of Fantasy Points and its Significance for National Football League Positional Groups

I. Introduction

The National Football League (NFL), established in 1920, quickly became the most popular and highest revenue-generating professional sport in the United States. Football is not only a fundamental part of American culture, but has also created a global presence by playing games in London, Mexico, Germany, and Brazil. In 2021, the NFL signed a record-setting 11-year media rights contract with CBS, NBC, Fox, ESPN, and Amazon worth over $100 billion. These media deals further expand the NFL's visibility and cement the league as a financial leader in the sports world (Belson). With the NFL's rise in popularity, fantasy football has grown in tandem, which gives fans a more active role in following their favorite players and the league in general. It is an integral part of the fan experience that blends management strategy and competition to enhance the NFL's impact both within the game and for the fanbase.

Fantasy football is a fun and engaging way for the average person to pick their ultimate NFL player dream team and compete against other drafted teams. Participants are able to join a league, made up of a minimum of ten participants, and draft and manage a roster of NFL players. The draft occurs before the first week of the NFL season and happens in a snake draft fashion. Each week, participants set a roster and compete head-to-head against another participant in the league. There are many different league settings that is unique to each fantasy platform but each team usually has "a weekly roster consisting of 12–16 spots and a starting lineup featuring a quarterback, one or two running backs, two wide receivers, a tight end, a flex (a running back or wide receiver), a kicker, and a team defense/special teams" (Zegura and Augustyn). The participants gain points by the actual performance of their rostered NFL players in their real life games.

Points can be calculated in different ways depending on league settings or platform. DraftKings has a scoring system, FanDuel has a scoring system, but the most common way to calculate points is through PPR, which stands for "points per reception." With this scoring system, players receive "1 point per 25 yards passing, 4 points per passing touchdown, -2 points per interception thrown, 1 point per reception, 1 point per 10 yards rushing/receiving, 6 points per TD, 2 points per two-point conversion, -2 points per fumble lost (est. prior to 1994)" (Pro Football Reference). There are many different fantasy football platforms with apps and websites that make participating easily accessible. Some of the popular ones are ESPN Fantasy Football, Yahoo Fantasy Football, NFL Fantasy Football, FanDuel Fantasy Football, and DraftKings Fantasy Football.

The legalization of sports gambling for the majority of states in the U.S. has led to an oversaturation of media and "professional" gambling advice where even major sports networks like ESPN have launched their own sportsbook, ESPN BET. With conflicting advice coming from all sides, it's difficult to discern credible sources from those who are not. Therefore, exploring the statistical properties of a gambling tool like fantasy football can provide a data-driven framework to make more informed decisions.

In this research paper, we will be analyzing the distribution of fantasy points scored in the years 2000-2023 based on the four offensive positions that form a fantasy starting lineup: quarterback (QB), running back (RB), wide receiver (WR), and tight end (TE). To compare the unique groups, we will calculate their coefficient of variation (CV) to determine each position's relative variability to the mean. This calculation allows for the comparison across groups with different means because it standardizes the measurement by including the sample's standard deviation. In general, a variable with a smaller CV has data that is less dispersed than a variable with a larger CV value (UCLA). The equation for CV is as follows:

$$\text{Coefficient of Variation (CV)} = \frac{\sigma}{\mu}$$

The CV for each position group will provide insight about the consistency of points scored by the players. Analyzing the CV for each position is important for making strategic decisions when it comes to draft night. This calculation will then help us determine which position is best to draft first. A theory we have is that the CV of wide receivers will be higher than the CV of quarterbacks because wide receivers only receive points when they are targeted and thrown to. There can be a wide range of receptions per wide receiver depending on the game tactics and strength of the opponents defense. Another theory is that tight ends will have the highest CV due to the inconsistency of talent in that position.

The hypotheses for our tests, using a significance level of 5%, are as follows:

Null Hypothesis: CV of all positions are the same

H_0: $\theta_{QB} = \theta_{RB} = \theta_{WR} = \theta_{TE}$

Alternative Hypothesis: CV of positions are different

H_A: $\theta_{QB} \neq \theta_{RB} \neq \theta_{WR} \neq \theta_{TE}$

The following report will be structured as follows: Data, Methods, Simulations, Analysis, and Discussion. The Data section provides introductory insight on the distribution of fantasy points scored by the four relevant positions for the sample data dating back to 2000, collected from ProFootballReference. The Methods section employs various methods to test the differing characteristics between the sample groups; Monte Carlo methods were used to generate samples from fitted probability distributions of PPR scores to estimate theoretical CV values for each position. Bootstrapping provided confidence intervals for the CVs and Jackknife methods tested the precision and variability estimates of these confidence intervals. The Analysis section describes and interprets the results and rejects or fails to reject the null hypothesis. The Analysis section also dives into a Bayesian analysis on predicting PPR for each player in each positional group. The Discussion section summarizes the report and provides context for future research.

II. Data

The data in this analysis was pulled directly from Pro Football Reference, a well-known and respected source for NFL statistics. The data includes the NFL fantasy rankings for every player in seasons 2000-2023 (Pro Football Reference). The data consists of the information in Pro Football Reference, ranging from historical and current player performance metrics, including statistics on teams and fantasy football. This dataset provides the most reliable possible ground for analyzing the differences between positions with respect to fantasy points. Twenty four datasets, one for each year, were downloaded from this webpage for the analysis and compiled into “fantasy_cleaned copy.csv." This large dataset has fantasy football data for every player who played in the NFL from 2000-2023. The dataset contains 14,283 observations (players) and 34 variables.

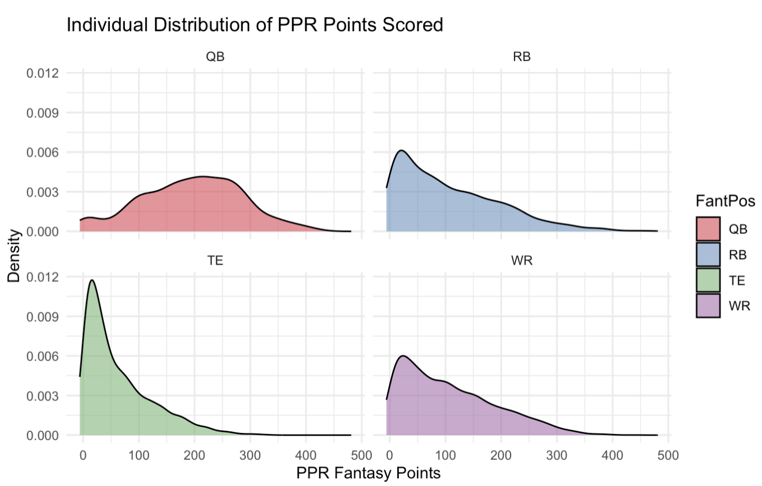
Relevant variables for the methods section are the following:

- Player - The name of the NFL player.
- FantPos - A player's position in fantasy football, such as QB for quarterback, RB for running back, WR for wide receiver, FB for fullback, and TE for tight end.
- PPR - Points Per Reception, the total fantasy football points scored.
- G - Number of games played.

For the methods section, these variables contain all of the information needed for the analysis. There is not a lot of data wrangling required as there are no missing entries for Player, FantPos, or G. There are some missing values for PPR so we filtered out all of the "NULL" values and values equal to zero. We also made the PPR value numeric to make analysis easier with limited bugs. We made sure every player name entry was unique and excluded the players who played fullback ("FB"). We did this because we are not interested in conducting an analysis on this positional group because of the limited number of true fullbacks in the NFL.

We filtered for only regular season data in the dataset to make the analyses both accurate and relevant. This is important because regular season games are the standard against which fantasy football scores are measured; pre-season or post-season games, for instance, might have different player dynamics, which would make them less comparable. We made games (G) an integer value and filtered out the players who participated in fewer than eight games within a season. We set an eight-game threshold, as seven games yielded 13,238 rows while eight reduced this to 9,631, offering a computationally efficient yet statistically significant dataset. This approach also excluded players with limited playing time due to injury or substitution.

Focusing on such variables, one will be in a position to develop an understanding of the distribution and variability of fantasy points (PPR) scored across four key fantasy football positions: quarterbacks, running backs, wide receivers, and tight ends. The analysis will point out the positions that normally have consistent performances. It will also show which positions have more variability, providing insightful information for those drafting teams in fantasy football. Below is a preliminary analysis of the true PPR points by position:



Pos	Min	Q1	Median	Mean	Q3	Max
QB	−3.70	131.45	200.50	194.50	259.90	417.40
RB	−4.10	30.50	84.65	107.43	164.93	481.10
TE	−1.40	16.50	41.80	63.30	93.70	330.90
WR	−6.00	36.20	90.10	106.75	160.60	439.50

Figure 1. Distribution of fantasy points, separated by position.

These outputs demonstrate that the distribution of fantasy points for RBs, WRs, and TEs are right-skewed with respective sample means of 106.75, 106.43, and 63.30. The distribution of QBs seems to be roughly symmetric, centered around 194.50 yearly fantasy points.

For further analysis, our group included a Bayesian analysis of PPR by player in each positional group for years 2018-2023. This work is included in the analysis section and relevant variables are the following:

- Player - The name of the NFL player.
- Age - Player's age on December 31st of that year.
- FantPos - A player's position in fantasy football, such as QB for quarterback, RB for running back, WR for wide receiver, FB for fullback, and TE for tight end.
- PPR - Points Per Reception, the total fantasy football points scored.
- G - Number of games played.
- GS - Number of games started.
- PosRank - The player's fantasy rank within his position.
- Year - Year the Fantasy Football season occurred

For the Bayesian analysis section, these variables contain all of the information needed for the analysis. Like above, there is not a lot of data cleaning required as there are no missing entries for Player, Age, FantPos, G, GS, PosRank, or Year. We first filtered the data for years 2018-2023 and made PPR, G, GS, Age, and PosRank numeric variables for easier data manipulation. We then fixed all of the players names by eliminating additional characters like "*" or "+" that Pro Football Reference uses to tag certain players. We did this to make sure players are not double counted or unaccounted for. We did not restrict the number of games played for this analysis because we are predicting the PPR for every individual player within each position for those years.

We then made 2023 into its own data frame to use as a testing data set for our predictive models. We found the intersection between our training dataset 2018-2022 and our testing dataset 2023, finding that there are 158 players who played in all six seasons. We then set all NA values equal to zero and removed four players who switched their position during our time frame. Grouping by position, we then made a training dataset of years 2018-2022, and testing dataset of year 2023, for QB, RB, WR, and TE. We did this so then we can make one model per positional group that predicts point totals for each player.

III. Method

We employed three different class methods in this research project: Monte Carlo simulations, Bootstrapping, and Jackknife analysis. These methods work together to provide a framework for evaluating CV or variability in PPR scores across positions. We start off with Monte Carlo which provides a theoretical baseline by generating random samples from fitted probability distributions other than approximating the behavior of a statistic, which in this case would be the CV. Bootstrapping is, “a class of nonparametric Monte Carlo methods that estimate the distribution of a population by resampling” (Rizzo 213). We used this by directly resampling the observed data to estimate coefficient variability (CV) and their confidence intervals (CIs) without relying on assumptions about distribution type. Finally, Jackknife was used to evaluate the performance of the bootstrap by measuring the precision of the CV estimates formed. Overall, these methods provided the best insights into the distribution and consistency of PPR scores relative to position using coefficient of variation as a metric. To understand more about these methods, refer to “Statistical Computing with “R by Maria Rizzo.

Before proceeding into procedures, assumptions, and theory, it is important to acknowledge the differences in data points for each position. As Figure 2 shows, there are four times more data points for WR than QB, which significantly influences variability estimates, particularly for methods like Jackknife and Bootstrap that are sensitive to the number of data points. This dispersion of data reasons out why we may see observed variability that contradicts what we would have expected to see. Ordinarily, the PPR scores for QBs are almost the same on a yearly basis, however, the smaller sample size increases the variability due to greater influence of outliers. On the other hand, the huge amount of data available for WR gives it the opposite effect as outliers will not significantly impact the variability, like Calvin Johnson’s 2011 season (Pro Football Reference). Rather than subsetting the data for each position to have the same sample size, we chose to use all available data to maximize the statistical power of the methods, capture positional nuances, and uphold the assumptions of bootstrap and jackknife, which require sufficiently large and representative datasets.

Number of Players by Position

FantPos	Count
QB	908
RB	2560
TE	2241
WR	3922

Figure 2. Number of players, separated by position.

<u>Monte Carlo Simulation</u>

Starting with Monte Carlo simulations, we used an "eye test" based on the density plot in Figure 1 to identify potential distributions by inspecting patterns like symmetry or skewness in the observed data. To test these assumptions, we used the 'fitdistr' function from the MASS package to fit the theoretical distributions to the observed data. After that, for each position a Kolmogorov-Smirnov (KS) test was conducted to evaluate a goodness of fit. Following the validation of the theoretical distributions, 10,000 Monte Carlo simulations were conducted for each position. This was done by drawing random samples from the fitted distribution and calculating the CV values for each simulation. These simulations were used to provide a distribution of estimates for the CV values for each position which would reflect the expected variability. This method assumed the known distribution for the PPR values in order to reflect the observed data and made use of the law of large numbers with the 10,000 simulations. It also ensured randomness in the simulation so that each iteration was statistically independent.

Bootstrapping

Using the information collected about distributions in Monte Carlo, we used Bootstrapping as a means to estimate the variability and confidence intervals (CIs) of CV values across the positions independently of their distribution. For each position, 1000 bootstrap samples were generated by resampling PPR scores with replacement, another assumption observed. Then, the CV value was calculated for each sample before finally the resulting distributions were aggregated to calculate the CIs for the CV of each position. This ensured that the resampling was conducted independently for each position. In order to evaluate if there was a significant difference in CV values between the positions, we used pairwise t-tests with Bonferroni corrections to control for multiple comparisons. This also helped us test our assumption that observations are independent of one another in further analysis.

Jackknife

We employed the Jackknife method to assess the precision of the variability estimates computed by the Bootstrap method, integrating a 5-fold cross-validation to balance computational efficiency and reliability. However as noted by Rizzo, “K-fold cross-validation partitions the data into subsets, where specific folds may disproportionately affect results due to outliers if not carefully managed”, potentially underestimating variability or missing fine patterns detectable with a leave-one-out approach. Our decision to include all the data mitigates this downside by maintaining a sufficiently large and representative dataset, which reduces the influence of statistically significant outliers and supports the assumption that the data is smooth and free from extreme outliers.

These three methods provided the framework for assessing the PPR scores of the positions in relation to their variability. These methods combine theoretical modeling, non parametric sampling, and sensitivity analysis to produce meaningful results. Further, the use of multiple methods and approaches allows us to have valuable insights for those interested in making strategic decisions for fantasy football.

IV. Simulations

In order to demonstrate the applicability of the methods above, we conducted simulations using data generated for the different positions. The simulations were performed under the assumptions that the observed data accurately follows the fitted distributions and that the assumptions are relaxed. The goal of these simulations is to observe the operating characteristics of the methods. This includes the CV and its sensitivity to deviations from the assumed distribution forms.

Monte Carlo Simulation

To understand the behavior of the data under the assumption that the data follows a known theoretical distribution, we generated 10,000 samples for each position within the fitted parameters. For each position, we calculated the CV in order to assess the variability in relation to the mean. Specifically:

- WR, RB, and TE samples were drawn from exponential distributions alongside their respective rate parameters.
- QB samples were drawn from a normal distribution parameterized by the fitted mean and standard deviation.

The results of the Monte Carlo simulations are summarized in this table below. The mean and standard deviation of the CV values were calculated to analyze the variability and stability under the assumptions.

Position	Assumed Distribution	Mean CV	SD of CV	Notes On Stability
WR	Exponential	1.0	0.10	Consistent to Assumptions
RB	Exponential	1.0	0.10	Consistent to Assumptions

TE	Exponential	1.0	0.10	Consistent to Assumptions
QB	Normal	0.5	0.04	Consistent and Stable under conditions.

Figure 3. Monte Carlo CV distribution by position

The density plot below shows consistent patterns across the positions. WR, RB, and TE have a mean CV of 1.0 and CV standard deviation (SD) of 0.1 which shows a stable variability across their simulations. The CV distribution for QB is narrower as it has a mean of 0.5 and an CV standard deviation (SD) of 0.04. These results are expected as the QB position introduces less variability to the exponential distribution. This supported the use of the normal distribution under our assumptions.

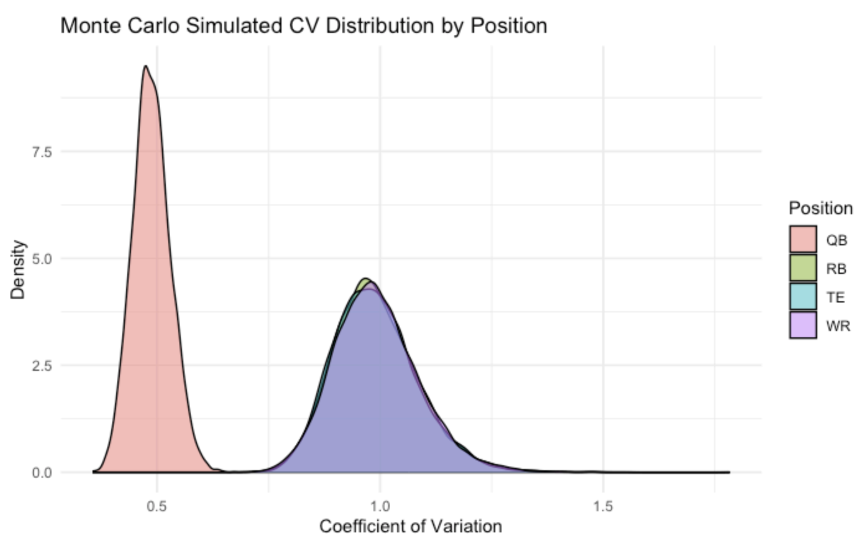


Figure 4. Distribution of CV from Monte Carlo methods, separated by position.

<u>Bootstrap Simulation</u>

We used bootstrap simulations for the synthetically generated data evaluating the precision and variability in estimates of CVs for our positions. Each data has been generated with n = 1000 observations using fitted exponential or normal distributions.

The bootstrap method was used to resample with replacement 1,000 times, and the CV was calculated for each bootstrap sample.

A pairwise t-test with Bonferroni correction was conducted to compare the CV distributions across these positions. Through the results, we can see that there are significant differences ($p < 2 \times 10^{-16}$) between all pairs:

Comparison	p-value
QB vs RB	< 2e - 16
QB vs TE	< 2e - 16
QB vs WR	< 2e - 16
RB vs TE	< 2e - 16
RB vs WR	< 2e - 16
TE vs WR	< 2e - 16

Figure 5. Pairwise t-test results

The 95% bootstrap confidence intervals for each position is as follows:

Position	Lower Bound	Upper Bound
WR	0.922	1.031
RB	0.946	1.050
TE	0.959	1.057
QB	0.311	0.340

Figure 6. Confidence Intervals from Bootstrap per position

The narrow intervals for QB showcases the lower variability we would expect due to its normal distribution assumption. On the other hand, the WR, RB, and TE positions show slightly wider intervals which align with their exponential distribution assumptions.

In the figure below, you can see the bootstrapped CV distributions by positions. Notably, the CV values for QB are concentrated around 0.3 which shows the low dispersion while WR, RB, and TE are around 1.0 with slightly more widespread distribution.

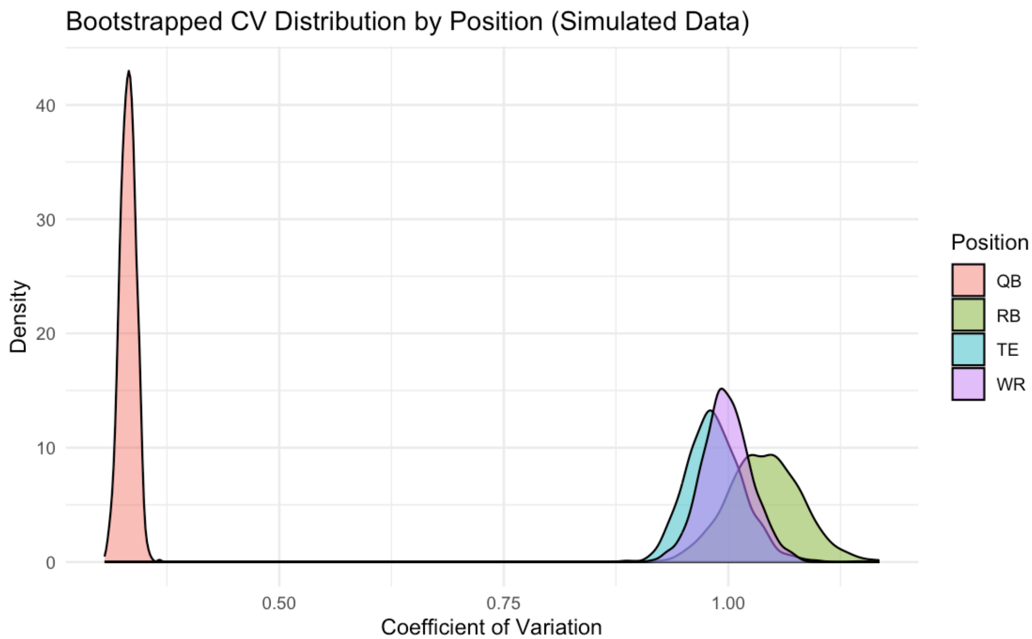


Figure 7. Bootstrapped CV distribution per position

<u>Jackknife Simulation</u>

For further investigation of the accuracy and variability of the CV estimates, we used the Jackknife resampling technique on the datasets for the positions. This technique removes an observation at a time and recalculates the CV in order to measure the variability of the estimates.

These are the variance of the Jackknife CV estimates for each position:

Position	Jackknife Variance
WR	2.68×10^{-4}
RB	9.99×10^{-5}
TE	9.69×10^{-4}

QB	7.80×10^{-5}

Figure 8. Jackknife Variance Per Position

As seen, TE had the highest Jackknife variance which means it has a higher dispersion and sensitivity for CV estimates. While QB had the lowest variance which meant it was tightly clustered with a lower variability as seen in the Bootstrap simulation analysis as well. The other two positions, WR and RB, had a moderate variance which also aligns them with their exponential distribution assumption. This dispersion can also be seen in the bar chart below:

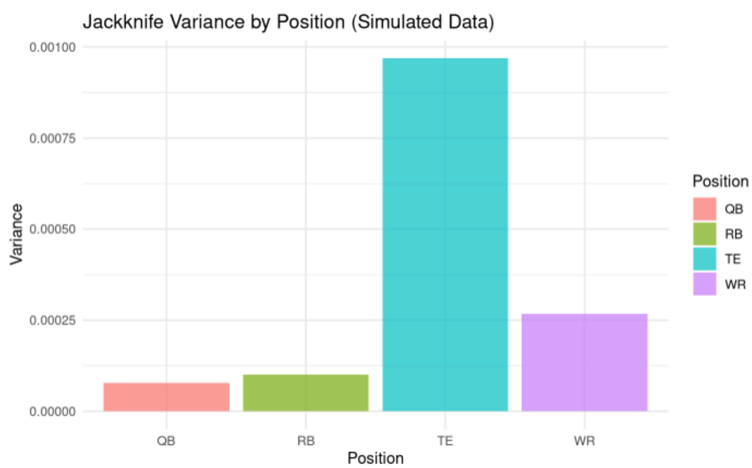


Figure 9.Bar Chart Distribution of Jackknife Variance

Kolmogorov-Smirnov (KS) test

Position	KS test value
WR	5.6×10^{-16}
RB	3.1×10^{-6}
TE	0.065
QB	0.124

Figure 10. Kolmogorov-Smirnoff test results

The results of the KS tests provide statistical evidence for the distributional assumptions across positions. The QB data fits a normal distribution well (p = 0.124), while the TE data shows marginal alignment with an exponential distribution (p = 0.065). In contrast, the WR (p = 5.6×10^{-16}) and RB (p = 3.1×10^{-6}) data do not fit the exponential distribution as closely. Given that the Monte Carlo simulations showed that we have confirmed there are two types of distributions for two different positions it makes sense to implement a Bootstrap to calculate CV by position for the reasons stated in the method and simulations.

V. Analysis

By the Pairwise Comparisons T-Test, Figure 13, we reject our null hypothesis that there are no significant differences in the means of the bootstrap-derived statistics between the compared groups. We uphold our assumption for each model that each observation is independent of one another; however, this assumption may not hold as we would expect that “QB vs TE” and “QB vs WR” would share some relationship as the PPR for the skill groups is somewhat dependent on the QB’s passing. However this may be negated as team relationships are lost due to the size of the dataset.

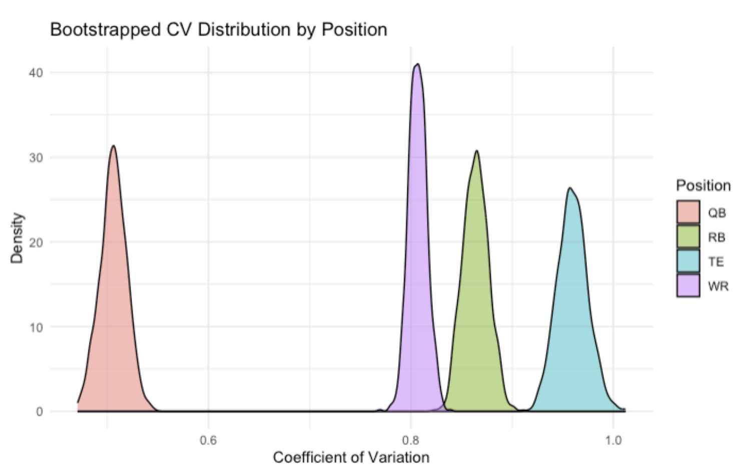


Figure 11. Bootstrapped CV Distribution by Position

Confidence Intervals for Bootstrap CV

Position	Lower Bound	Upper Bound
WR	0.7888301	0.8254385
RB	0.8413315	0.8900748
TE	0.9301911	0.9880605
QB	0.4800996	0.5290510

Figure 12. Confidence Intervals for Bootstrap CV

Pairwise Comparisons Using T-Tests

Comparison	P-Value
QB vs RB	< 2e-16
QB vs TE	< 2e-16
QB vs WR	< 2e-16
RB vs TE	< 2e-16
RB vs WR	< 2e-16
TE vs WR	< 2e-16

Figure 13. Pairwise Comparisons Using T-Tests

The Bootstrapped CV values in Figure 11 reveals the relative variability of player performance across positions. QBs have the lowest CV, indicating the most consistent performance and the least relative variability. This means that no matter which quarterback you draft, they will probably score a similar amount. Then followed by WRs and RBs which means there are a few more standout players in each position that account for this difference in variability. This signifies that you could use an earlier pick on a standout WR or RB. Lastly, TEs have the highest CV, showing the greatest relative variability and making them the least predictable. These CV values emphasize the significant positional differences in player performance, with QBs being the most reliable and TEs the most variable.

The bootstrap confidence intervals in Figure 12 reveal important differences in the precision of these CV values. WRs exhibit the narrowest confidence interval (0.7888, 0.8254), indicating the most precise estimates. Followed by RBs and QBs with slightly wider intervals ((0.8413, 0.8901) and (0.4801, 0.5291), respectively). TEs, however, have the largest interval (0.9302, 0.9881), indicating the least precision and the highest relative variability.

5-Fold Variance of Bootstrap CV by Position	
Position	5-Fold Variance
QB	2.282446e-05
RB	4.337427e-05
TE	1.246882e-04
WR	3.188521e-05

Figure 14. 5-Fold Variance of Bootstrap CV by Position

The jackknife method assesses the precision of bootstrap estimates by measuring the variability when each observation is systematically removed, using 5-fold variances of bootstrap CV. The results in Figure 14 indicate that QBs have the smallest variance or greatest precision (2.28×10^{-5}), suggesting consistent bootstrap CV estimates. RBs and WRs show slightly larger variances (4.34×10^{-5} and 3.19×10^{-5}, respectively), reflecting moderate precision. TEs exhibit the highest variance (1.25×10^{-4}), suggesting they are the least precise bootstrap estimates. This aligns with the advice of many fantasy football guides, which recommend drafting quarterbacks in later rounds despite their high average point production due to their consistent PPR regardless of player. In contrast, the greater variability and lower precision of TE performance explain why sports blogs often encourage players to prioritize drafting elite TEs like Travis Kelce first overall, as his consistent high-scoring performance can provide a significant advantage in a less predictable position (Fantasy Sharks). If you get the best tight end, you will be in great Fantasy Football shape because there is a large dropoff of point total from the best to the second best. This is supported by the large variance.

Interestingly, QBs have both the smallest jackknife variance and the lowest CV, yet not the smallest bootstrap CI. This violates our assumption that all three should match in order of magnitude across position. However, with four times fewer QBs than WRs in the data, the smaller QB sample size likely reduces resampling variability, leading to a smaller jackknife variance. At the same time, WRs' larger sample size supports a narrower CI despite their slightly higher jackknife variance. This highlights WRs as a reliable position, benefiting from greater data representation, while QBs retain strong precision due to their inherently structured roles.

Predicting PPR Points with Bayesian Modeling

To further explore the alternative hypothesis, we fit a Bayesian multivariate model for each of the four positions to predict fantasy points for the 2023 season using five years of data from the 2018 to 2022 seasons.

We wanted to fit hierarchical Bayesian linear regression models using stan and did so by using the "brms" package in R studio. This package "implements Bayesian multilevel models in R using the probabilistic programming language Stan" (Bürkner) and we ran all of our models with brm. This brm model is great for complex data and allowed us flexibility with establishing our family specific parameters (Bürkner). It also calls stan and uses NUTS sampler for credibility.

This can illustrate the advantages of selecting wide receivers by using posterior predictions, which reinforces the results found when analyzing the coefficient of variation data. Using the individual distributions of fantasy points in Figure 1 and the Monte Carlo simulation results, we assume that the data for quarterbacks follows a Guassian (normal) distribution and the data for running backs, wide receivers, and tight ends follow a Skewed-Normal distribution. Given our implementation of Monte Carlo simulation to determine the distribution of PPR points for each position, we originally planned to model the tight ends using a Gamma family with a log link function; however, this was not possible due to some players with negative PPR points. Additionally, the models utilize default, uninformative priors so that the posterior distribution closely reflects the distribution of the data. For each of the four models, four chains were used, and each chain was run for 4,000 iterations after a burn-in period of 2,000 iterations; standard diagnostic plots about convergence indicated that this is sufficient (Raftery). The general Bayesian model for each position is defined below:

$$\mu_i = \beta_0 + u_{Player} + \beta_{Age}Age_i + \beta_G G_i + \beta_{GS} GS_i + \beta_{PosRank} PosRank_i$$

We use a random effect specification on the Player variable, represented by u_{Player}. This allows for the estimation of a different intercept for each player to capture unique player trends. Multiple sets of response variables were tested against each other to choose the ideal model using leave-one out cross validation from the Loo package. Ultimately, the set of variables in the model equation above is preferred because it yields the highest expected log density and thus, were used for each positional model.

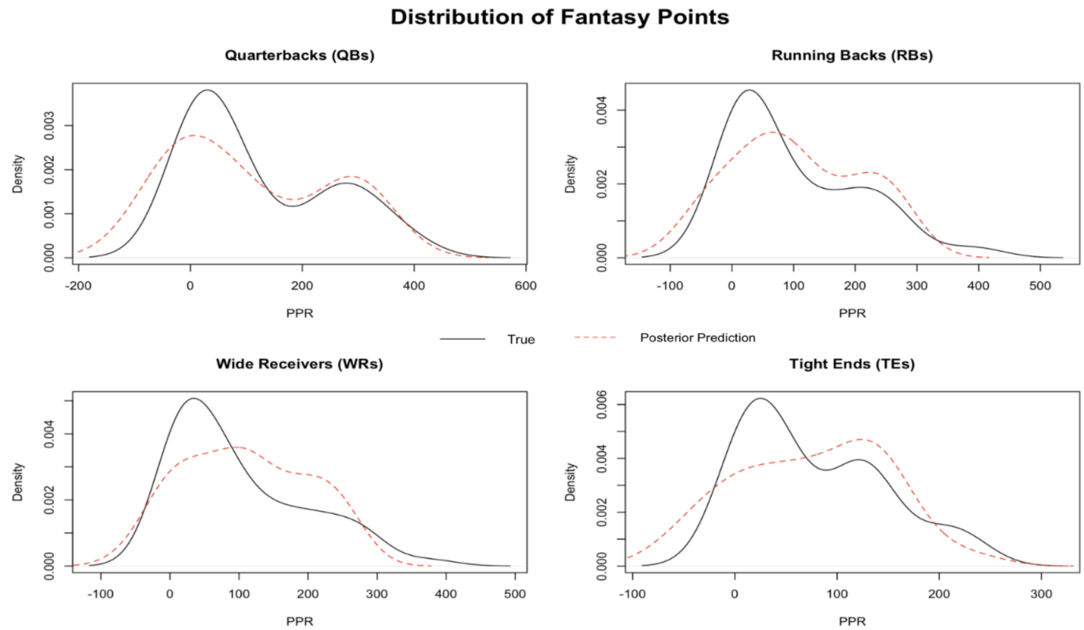
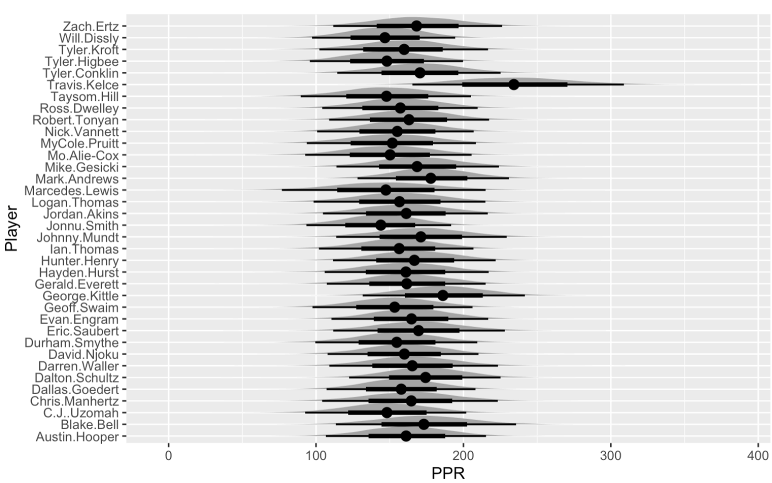


Figure 15. *Distribution of true PPR points scored by position in 2023 compared to our posterior predictions.*

From our models, we found that the posterior mean of the quarterback group for PPR points is 109.79, with a 95% credible interval of (-78.38, 324.12). This suggests that on average, we expect a quarterback from the 2023 season to score approximately 109.79 points, which is the largest posterior mean of the four groups. The wide range of a confidence interval suggests that there is a significant level of variability in the predictions, reflected by the differences in values of the predictor variables for each player. The confidence intervals for the other positions are similarly uncertain, but are narrower compared to the quarterbacks. Running backs have a posterior mean of 104.56 with a 95% credible interval of (-54.82, 269.49), the wide receivers have a posterior mean of 109.65 with a 95% credible interval of (-28.82, 260.09), and the tight ends have the smallest posterior mean, 75.59, with a 95% credible interval of (-45.05, 195.13). To further evaluate the performance of our predictive models, we calculate and attempt to minimize the loss of each model. We utilize Mean Squared Error (MSE) to measure the amount of errors in our model. The standard MSE formula,

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

compares the true value for PPR points scored by each position in 2023 to our posterior predictions, penalizing larger errors more heavily. The data for quarterbacks, running backs, and wide receivers yield similar loss values of 1112.79, 1343.91, 1110.87, respectively while our model for tight ends has a significantly lower loss value of 669.11. Although the tight end group yielded the largest CV value using Bootstrapping and Jackknife methods, the year-to-year consistency of individual tight ends makes them easy to model when incorporating the random effect specification on the “Player” variable.



Top 10 TEs in 2023 (by PPR)

Player	PPR	PPR_Predicted	Predicted_Lower	Predicted_Higher
George Kittle	203.2	189.5186	144.15937	255.3701
Travis Kelce	219.4	234.4415	170.23337	306.8390
David Njoku	201.2	158.9137	114.63578	223.8436
Evan Engram	230.3	159.1575	115.38012	224.4080
Taysom Hill	143.5	128.8484	85.93166	194.9425
Dalton Schultz	150.5	152.6377	107.70121	218.1690
Mark Andrews	135.4	158.1703	114.10738	224.9904
Hunter Henry	119.9	142.3733	99.08164	207.4081
Dallas Goedert	136.3	137.4418	93.68307	201.4874
Jonnu Smith	124.2	107.2567	61.00113	175.6157

Figure 16. Posterior densities of PPR points for tight ends with 95% confidence intervals

The output above further demonstrates the variability of the predicted points of each player for the tight end group. On average, tight ends score the least amount of points when it comes to comparing positional groups because they are usually just blockers. Except for the outlier performance of Travis Kelce, who was identified as a standout player in our Jackknife methods. This might be attributed to a few outlier players who are classified as tight ends, but perform more closely to a wide receiver because they do not have blocking responsibilities on their teams.

For each position, we compared the 95% posterior credible intervals of PPR points for each player to their true number of points scored to measure the coverage rate. All quarterback's points scored in 2023 fall within our model's 95% credible interval, 96.67% of the true data for running backs fall within the 95% interval, 94.55% of the true data for wide receivers fall within the 95% interval, and 94.44% of the true data for tight ends fall with the 95% interval using our model. The results demonstrate that each of our four models provide high coverage rates and accurately define a credible interval for evaluating player outcomes.

Using 1000 replications with replacement, we simulate taking a random player sample from the posterior predictions to find the probability that selecting one position is better than another. It is important to note that these samples come from only the players who played every season from 2018 to 2023. These results suggest that the probability that a randomly selected quarterback scores more PPR points than a randomly selected running back or wide receiver is .48 and .471, respectively; however, the probability that a randomly selected quarterback scores more PPR points than a randomly selected tight end is .506. This signifies that a quarterback is likely a better pick than a tight end, but is a worse pick earlier in a fantasy draft when compared to running backs or wide receivers. Similarly, randomly selecting a tight end from 1000 simulations always yields a smaller PPR value; the likelihood that a tight end scores more than a quarterback is .494, the likelihood of scoring more than a running back is .431, and the likelihood of scoring more than wide receiver is .418. Of the four positions, wide receivers are likely to be a better selection when compared to any of the other groups.

	QB	RB	WR	TE
QB	-	0.48	0.471	0.506
RB	0.52	-	0.477	0.569
WR	0.529	0.523	-	0.582
TE	0.494	0.431	0.418	-

Figure 17. Probability that a position (Left Column) scores more than an different position (Top Row)

VI. Discussion

In conclusion, this paper analyzed the coefficient of variation (CV), represented by θ, and its statistical properties for each of the four positions integral to fantasy football. The hypothesis tested that the CV of quarterbacks, running backs, wide receivers, and tight ends were all the same, $\theta_{QB} = \theta_{RB} = \theta_{WR} = \theta_{TE}$, was rejected. The alternative was accepted and explored the significance of differing values of θ and its application to the strategy of fantasy football.

Using exploratory analysis of the distributions of PPR for each position paired with the results from Monte Carlo testing, there was strong evidence to suggest that the quarterback position followed a normal distribution whereas the tight end group followed an exponential distribution; however, there was not strong enough evidence to comment on the distribution of the running back and wide receiver groups.

The bootstrapped CV values highlight quarterbacks as the most consistent and least variable, followed by wide receivers and running backs, with tight ends being the most variable and least predictable. The bootstrap confidence intervals show wide receivers have the most precise estimates, while jackknife results confirm quarterbacks have the smallest variance, reflecting high precision despite their slightly wider CI compared to wide receivers. This discrepancy may be explained by the dataset composition, where fewer quarterbacks reduce resampling variability, giving quarterbacks strong precision, while wide receivers' larger sample size supports their narrower CI, underscoring the reliability of both positions for different reasons.

In the Analysis section, we use our results from the previous methods as a baseline to set up Bayesian predictive models for each position. Our results illustrate that wide receivers are the most advantageous pick because of their high posterior mean predictions combined with larger variance; this makes outlier players more likely when compared to other positions like the quarterback group. In contrast, the tight end group had the lowest probability of outperforming the other positions; however, it is important to note that the tight end model yielded the highest predictive accuracy because of the consistency of tight end specific trends.

Our research provides practical advice about the tradeoff between stable positions that offer a higher floor of PPR points, like quarterbacks, and riskier positions that offer a higher ceiling because of their variability, like wide receivers. Using this, players can make informed decisions about balancing stable and high risk/reward players to maximize their rosters.

Our detailed analysis is just the beginning for understanding the differences in fantasy football PPR scored across different positions. We investigated quarterbacks, wide receivers, running backs, and tight ends to understand the different scoring trends of each position. Looking across all of our methods, our analysis shows that a great draft strategy is to target standout WRs in the first or second round to balance both a low CV and a high point total. Then, prioritizing positions with a lower CV value, or more consistency, earlier in the draft, like QBs over TEs. Then, draft players with a higher CV, or higher risk, higher reward, positions, like bottom of the barrel tight ends, later in the draft. Drafting positions with more consistency could lead to a greater payout in the end.

This is a great foundation for further analysis of how to make strategic data driven decisions when it comes to drafting strategies for fantasy football. Advanced analysis could simulate different draft strategies and see which accumulates the highest point total during a season. Future research could include more positions that are included in fantasy football scoring like fullbacks, kickers, and defenses. Investigating the impact of different positions and how they are valued in different leagues and scoring statistics could also be analyzed. We faced limitations when it came to excluding players with few games, future research could use every player for a more realistic analysis. We also should have excluded players who faced injuries during the season because that could have skewed the data. Instead, we could have made a separate analysis of how players perform the year following injury. This could sway how fantasy football participants draft players who are historically injured. More advanced statistics like individual player yard averages or catch probability could also be added to the Bayesian analysis for better predictive models. External variables like weather and strength of schedule could also refine the analysis. This detailed analysis can also be expanded to other fantasy sports as taking past data is a great indicator for future performance.

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VIII. Supplemental Materials

1. R-Markdown file submitted with all analysis for this paper in a single pass.
2. CSV file of fantasy football data is uploaded to Canvas.